

open-questions-ai

Mohan Chinnappan
mohan.chinnappan.n@gmail.com

June 2026

Abstract

Modern large language models (LLMs) achieve remarkable performance on a broad range of tasks, yet we lack a principled theory explaining *why* they reason, generalise, or hallucinate the way they do. This paper surveys five grand open questions at the frontier of AI research: (1) **mechanistic interpretability** — can we reverse-engineer neural networks into human-understandable circuits?; (2) **world models and causality** — do LLMs learn implicit causal models, and can explicit world models do better?; (3) **reasoning and computational depth** — are transformer reasoning abilities genuine, and what are their fundamental limits?; (4) **representation collapse and JEPA** — does predicting in latent space rather than signal space produce more structured, interpretable representations?; and (5) **modular AI** — will future general systems be monolithic or composed of specialised coordinated modules? For each question we provide precise mathematical formulations, discuss the current state of knowledge, and illustrate key concepts with PyTorch code. We argue that answering these questions requires not only scaling, but fundamentally new architectural inductive biases.

Keywords: interpretability, world models, JEPA, mechanistic interpretability, causal reasoning, sparse autoencoders, self-supervised learning, energy-based models.

Introduction

The past five years have witnessed extraordinary advances in AI: LLMs such as GPT-4, Gemini, and Claude solve complex reasoning problems, write code, and pass professional examinations. Despite this empirical success, we lack a complete theoretical understanding of *why these systems work* and *where they fundamentally fail*.

This gap has produced entire research sub-fields: *mechanistic interpretability* attempts to reverse-engineer neural network circuits; *representation learning* studies the geometry of learned feature spaces; *world model research* asks whether models build causal models of their environments; and *scaling law research* asks whether the current paradigm can reach general intelligence.

We organise the open questions as follows. Section 3 covers mechanistic interpretability. Section 4 covers world models and causality. Section 5 covers reasoning and computational limits. Section 6 covers the JEPA architecture as a

concrete alternative inductive bias. Section 7 discusses the modular AI hypothesis. Section 9 concludes with an integrated research agenda.

Background: What an LLM Computes

An autoregressive language model parameterised by θ computes

$$P_{\theta}(y_1, \dots, y_T) = \prod_{t=1}^T P_{\theta}(y_t \mid y_{<t})$$

where $y_{<t} = (y_1, \dots, y_{t-1})$ is the context. Training minimises the negative log-likelihood:

$$\mathcal{L}_{\text{NLL}}(\theta) = -\frac{1}{T} \sum_{t=1}^T \log P_{\theta}(y_t \mid y_{<t}).$$

A transformer implements this through L alternating self-attention and feed-forward layers. Each self-attention head computes

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V,$$

where $Q = hW_Q$, $K = hW_K$, $V = hW_V$ are linear projections of the hidden state $h \in \mathbb{R}^{d_{\text{model}}}$. The full residual-stream update is

$$\begin{aligned} h_{\ell} &= \text{LayerNorm}(h_{\ell-1} + \text{MHA}(h_{\ell-1})) \\ h'_{\ell} &= \text{LayerNorm}(h_{\ell} + \text{FFN}(h_{\ell})). \end{aligned}$$

Critically, Equation [eq:nll] contains no term that rewards physical understanding, causal reasoning, or goal-directed planning. Any such capability must *emerge* from next-token prediction alone — and this is the root cause of all five open questions.

PyTorch: Basic Transformer Block

```
import torch, torch.nn as nn, math

class TransformerBlock(nn.Module):
    def __init__(self, d_model, n_heads, d_ff):
        super().__init__()
        self.attn = nn.MultiheadAttention(
            d_model, n_heads, batch_first=True)
        self.ff = nn.Sequential(
            nn.Linear(d_model, d_ff),
            nn.GELU(),
            nn.Linear(d_ff, d_model))
        self.ln1 = nn.LayerNorm(d_model)
        self.ln2 = nn.LayerNorm(d_model)
```

```

def forward(self, x):
    # x: (B, T, d_model)
    h = self.ln1(x)
    h, _ = self.attn(h, h, h)
    x = x + h # Eq. (3)
    x = x + self.ff(self.ln2(x)) # Eq. (4)
    return x

```

Open Question 1: Mechanistic Interpretability

Definition 1 (Mechanistic Interpretability). *The program of mechanistic interpretability aims to identify human-understandable algorithms implemented by the weights of a neural network — analogous to reverse-engineering a compiled binary into readable source code .*

The Superposition Hypothesis

A key obstacle is *polysemanticity*: single neurons respond to multiple, semantically unrelated concepts. propose the **superposition hypothesis**: a network of dimension d can represent up to $n \gg d$ features by encoding them as near-orthogonal directions in $\mathbb{R}^d$, at the cost of inter-feature interference.

Proposition 1 (Superposition Capacity, informal). *If features are sparse — each is active with probability $p \ll 1$ — then $n \approx d \cdot (-\log p)^{-1}$ features can be stored with interference below a threshold ϵ .*

The activation of a polysemantic neuron j for input $\mathbf{x}$ is

$$h_j = \text{ReLU}\left(\sum_i W_{ji}x_i + b_j\right),$$

where multiple features i share weight W_{ji} , making clean attribution impossible.

Sparse Autoencoders

Sparse autoencoders (SAEs) decompose polysemantic activations into monosemantic features by solving

$$\begin{aligned}
\mathbf{z} &= \text{ReLU}(W_e \mathbf{h} + \mathbf{b}_e) \\
\hat{\mathbf{h}} &= W_d \mathbf{z} + \mathbf{b}_d \\
\mathcal{L}_{\text{SAE}} &= \|\mathbf{h} - \hat{\mathbf{h}}\|_2^2 + \lambda \|\mathbf{z}\|_1,
\end{aligned}$$

where $W_d \in \mathbb{R}^{d_{\text{model}} \times d_{\text{SAE}}}$ with $d_{\text{SAE}} \gg d_{\text{model}}$. Each column of W_d is a candidate *feature direction*.

```
class SparseAutoencoder(nn.Module):
    def __init__(self, d_model, d_sae):
        super().__init__()
        self.W_enc = nn.Linear(d_model, d_sae)
        self.W_dec = nn.Linear(d_sae, d_model,
                                bias=False)
        self.b_enc = nn.Parameter(
            torch.zeros(d_sae))

    def forward(self, h):
        # Eq. (5)
        z = torch.relu(self.W_enc(h) + self.b_enc)
        h_hat = self.W_dec(z)          # Eq. (6)
        return h_hat, z

def sae_loss(h, h_hat, z, lam=1e-3):
    recon = (h - h_hat).pow(2).mean() # Eq. (7)
    sparse = lam * z.abs().mean()
    return recon + sparse
```

Linear Representation Hypothesis

The **linear representation hypothesis** posits that concepts are encoded as linear directions in activation space. Formally, concept c is encoded at layer ℓ if there exists $\mathbf{v}_c \in \mathbb{R}^d$ such that

$$\hat{c}(\mathbf{h}_\ell) = \text{sign}(\mathbf{v}_c^\top \mathbf{h}_\ell - \theta)$$

achieves high accuracy. Causal evidence requires that patching

$$\mathbf{h}_\ell \leftarrow \mathbf{h}_\ell + \alpha \mathbf{v}_c$$

changes the model’s output in the direction predicted by concept c .

The Scaling Barrier

Remark 1. *Despite impressive progress, mechanistic interpretability faces a fundamental scaling barrier: manually annotating circuits for a model with $\sim 10^{12}$ parameters is computationally infeasible. Automated methods (SAEs, probing) cover only a small fraction of model behaviour, and there is no ground-truth evaluation criterion for whether an interpretation is “correct.”*

Open Question 2: World Models and Causality

Definition 2 (World Model). A world model is a parametric function $f_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ that predicts future states from current states and actions, trained to minimise

$$\mathcal{L}_{\text{WM}}(\theta) = \mathbb{E}_{(s_t, a_t, s_{t+1})} [\|f_\theta(s_t, a_t) - s_{t+1}\|_2^2].$$

The Blurry Prediction Problem

When the target s_{t+1} is stochastic and the prediction is made directly in signal space (pixels, tokens), the optimal predictor is the conditional mean

$$\hat{s}_{t+1}^* = \mathbb{E}[s_{t+1} \mid s_t, a_t],$$

which is a blurry average of all possible futures. If a ball can move left or right with equal probability, the optimal $\hat{s}^*$ is a ghost ball at the centre — physically impossible.

Causal Reasoning

Pearl's **do-calculus** distinguishes observational from interventional distributions:

$$\begin{aligned} \text{Observational: } & P(Y \mid X = x) \\ \text{Interventional: } & P(Y \mid \text{do}(X = x)) \\ \text{Counterfactual: } & P(Y_{X=x} \mid X = x', Y = y'). \end{aligned}$$

LLMs trained on observational text (Equation [eq:obs]) cannot in general compute interventional or counterfactual quantities (Equations [eq:int]–[eq:counter]) without explicit causal structure.

```
class WorldModel(nn.Module):
    def __init__(self, d_state, d_action, d_h):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(d_state + d_action, d_h),
            nn.SiLU(),
            nn.Linear(d_h, d_h),
            nn.SiLU(),
            nn.Linear(d_h, d_state))

    def forward(self, s, a):
        # s: (B, d_state), a: (B, d_action)
        return self.net(torch.cat([s, a], -1))

# Eq. (8): MSE loss in latent space
loss_fn = nn.MSELoss()
```

Emergent World Models in LLMs

showed that a transformer trained on Othello move sequences develops an internal representation of the board state that can be recovered by linear probing. This suggests LLMs do develop *implicit* world models — but whether these are genuinely causal or merely correlational is an open question.

Open Question 3: Reasoning and Computational Depth

Transformers as Bounded-Depth Circuits

A transformer with L layers computes a function of fixed depth:

$$f_\theta = f_L \circ f_{L-1} \circ \cdots \circ f_1, \quad \text{depth}(f_\theta) = O(L).$$

proved that log-precision transformers cannot solve problems outside TC^0 in a single forward pass. Problems requiring $\omega(L)$ sequential steps therefore cannot be solved by a single forward pass.

Chain-of-Thought as External Memory

Generating T additional output tokens extends effective depth to $O(L \cdot T)$, explaining why chain-of-thought (CoT) prompting dramatically improves multi-step reasoning.

The per-token log probability under CoT is

$$\log P_\theta(y_t \mid r_1, \dots, r_k, y_{<t})$$

where $r_1, \dots, r_k$ are reasoning tokens. The scratchpad allows the model to re-read intermediate computations as context.

Scaling Laws

derived compute-optimal scaling laws:

$$\mathcal{L}(N, D) \approx E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$$

where N is parameter count, D is data tokens, and $\alpha, \beta \approx 0.34$. The irreducible loss E represents the entropy of natural language.

Remark 2. *Emergent abilities appear as sharp transitions in performance at certain scale thresholds. Whether these are genuine phase transitions or artifacts of the evaluation metric is debated.*

Open Question 4: JEPA and Representation Space Prediction

The JEPA Objective

The **Joint Embedding Predictive Architecture** (JEPA) replaces next-token prediction with prediction in latent representation space. Given context x and target y , JEPA minimises

$$\begin{aligned} s_x &= E_x(x), \quad s_y = E_y(y) \\ \hat{s}_y &= g(s_x, z) \\ \mathcal{L}_{\text{JEPA}} &= \|g(s_x, z) - \text{sg}(s_y)\|_2^2, \end{aligned}$$

where $\text{sg}(\cdot)$ is the stop-gradient operator, E_y is an exponential moving average (EMA) of E_x , and z is a latent variable encoding information about y not present in x .

EMA Update and Collapse Prevention

The EMA update

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau)\theta, \quad \tau \in [0.996, 1.0)$$

provides a slowly-evolving, stable prediction target, preventing representational collapse without requiring contrastive negative pairs.

```
import copy
```

```
class JEPA(nn.Module):
    def __init__(self, enc, pred, tau=0.996):
        super().__init__()
        self.enc_x = enc
        self.enc_y = copy.deepcopy(enc)
        for p in self.enc_y.parameters():
            p.requires_grad_(False)
        self.predictor = pred
        self.tau = tau

    @torch.no_grad()
    def ema_update(self):
        # Eq. (14)
        for po, pe in zip(
            self.enc_x.parameters(),
            self.enc_y.parameters()):
            pe.data.mul_(self.tau).add_(
                po.data, alpha=1 - self.tau)

    def forward(self, x, y, z=None):
```

```

s_x = self.enc_x(x) # Eq. (12)
with torch.no_grad():
    s_y = self.enc_y(y) # Eq. (12)
s_hat = (self.predictor(s_x, z)
         if z is not None
         else self.predictor(s_x))
return (s_hat - s_y.detach()
        ).pow(2).mean() # Eq. (13)

```

VICReg: Preventing Collapse via Regularisation

VICReg prevents collapse without negative pairs through three explicit regularisation terms:

$$\mathcal{L}_{\text{VICReg}} = \lambda \underbrace{\|z_1 - z_2\|_2^2}_{\text{invariance}} + \mu \underbrace{v(z_1, z_2)}_{\text{variance}} + \nu \underbrace{c(z_1, z_2)}_{\text{covariance}}$$

where $v(z) = \sum_j \max(0, \gamma - \text{Std}(z_j))$ and $c(z) = \sum_{i \neq j} [C(z)]_{ij}^2 / d$ with $C(z) = z^\top z / (N - 1)$ the normalised covariance matrix.

H-JEPA: Hierarchical World Model

The **Hierarchical JEPA** stacks K JEPA levels at increasing temporal scale and abstraction:

$$\begin{aligned}
s_t^{(k)} &= \text{Pool}(s_t^{(k-1)}), \quad k = 2, \dots, K \\
\hat{s}_{t+1}^{(k)} &= g^{(k)}(s_t^{(k)}, a_t^{(k)}, z^{(k)}) \\
\mathcal{L}^{(k)} &= \|\hat{s}_{t+1}^{(k)} - \text{sg}(s_{t+1}^{(k)})\|_2^2.
\end{aligned}$$

Level $k = 1$ makes fine-grained short-horizon predictions; level $k = K$ makes abstract long-horizon predictions suitable for high-level planning.

What JEPA Does Not Solve

Remark 3. *JEPA substantially improves world model quality and enables gradient-based planning (Section 7). However, it does not solve interpretability: a latent vector $[0.34, -1.82, 0.77, \dots] \in \mathbb{R}^d$ is no more interpretable than an LLM hidden state. The learned latent space may be more structured, but it is still distributed and not symbolically grounded.*

Open Question 5: Modular AI and Gradient-Based Planning

The Six-Module Architecture

proposes replacing a monolithic LLM with six specialised modules: Configurator, Perception, World Model (H-JEPA), Short-Term Memory, Cost Module, and Actor. The Actor performs **inference-by-optimisation**:

$$a^* = \arg \min_{a_{0:H}} \sum_{t=0}^H c(s_t, a_t), \quad s_{t+1} = f_\theta(s_t, a_t),$$

where the world model f_θ is fully differentiable, so $\nabla_{a_{0:H}} \mathcal{L}$ can be computed by backpropagation — planning without real-world trial and error.

```
def plan(world_model, cost_fn, s0,
        horizon=10, n_steps=100, lr=1e-2):
    """
    Solve Eq. (18) by gradient descent.
    world_model: (s, a) -> s_next
    cost_fn:      s      -> scalar
    """
    d_a = s0.shape[-1] # action dim = state dim
    actions = torch.zeros(
        horizon, d_a, requires_grad=True)
    opt = torch.optim.Adam([actions], lr=lr)
    for _ in range(n_steps):
        s, total_cost = s0, 0.0
        for t in range(horizon):
            s = world_model(s, actions[t])
            total_cost = total_cost + cost_fn(s)
        opt.zero_grad()
        total_cost.backward() # grads thru WM
        opt.step()
    return actions.detach()
```

Information Efficiency of Self-Supervised Learning

Self-supervised learning provides approximately 10^5 bits of training signal per sample (one bit per masked patch), compared to ~ 10 bits per sample for supervised learning (one class label). This $10^4 \times$ advantage makes SSL the natural training paradigm for world models.

Comparison of Approaches

Table 1 compares architectures across the five open questions.

Architecture comparison across open questions. = addressed, $\sim$ = partially addressed, $\times$ = unsolved.

Architecture	Interp.	WM	Causal	Plan	Halluc.
--------------	---------	----	--------	------	---------

Architecture	Interp.	WM	Causal	Plan	Halluc.
LLM (GPT)	×	~	×	~	×
LLM + RLHF	×	~	×	~	~
I-JEPA	~	~	×	×	N/A
H-JEPA	~		~		N/A
H-JEPA+Modules	~		~		~

An Integrated Research Agenda

Drawing together the five open questions, we identify a set of high-priority research directions:

1. **Automated circuit discovery at scale.** Scaling SAEs and automated probing to 10^{12} -parameter models requires new algorithmic efficiency — the bottleneck is not expressiveness but compute.
2. **Causal inductive biases.** Training objectives that explicitly reward interventional reasoning (Equation [eq:int]), rather than purely observational next-token prediction (Equation [eq:lm]).
3. **Structured latent spaces.** Designing latent spaces with explicit structural priors — e.g., factored representations, object-centric slots — so that semantic dimensions are interpretable by construction.
4. **Hierarchical world models.** Validating H-JEPA on embodied planning tasks that require long-horizon reasoning across multiple abstraction levels.
5. **Evaluation benchmarks.** Ground-truth evaluation criteria for interpretability claims, causal model quality, and planning performance in open-ended environments.

Conclusion

We have surveyed five grand open questions at the frontier of AI research. Current LLMs achieve remarkable performance but lack interpretable circuits, explicit causal models, and robust planning capabilities. Architectures such as JEPA and H-JEPA address the world model and planning questions but do not resolve interpretability. Modular systems combining explicit world models, memory, and gradient-based planners represent a promising direction, but require new training algorithms, evaluation methods, and theoretical foundations.

The central message is that answering these questions requires not only scaling existing architectures, but fundamentally reconsidering the inductive biases of learning systems — moving from pure next-token prediction toward architectures that explicitly represent states, transitions, and causal structure.

Acknowledgements

The author thanks the mechanistic interpretability and self-supervised learning communities for making their code and models publicly available.

99

Anthropic (2024). The Claude 3 model family: Opus, Sonnet, Haiku. *Technical Report*, Anthropic.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. (2023). Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR 2023*.

Bardes, A., Ponce, J., and LeCun, Y. (2022). : Variance-invariance-covariance regularization for self-supervised learning. In *ICLR 2022*.

Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., et al. (2023). Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*.

Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., et al. (2022). Toy models of superposition. *Transformer Circuits Thread*.

Gemini Team (2023). Gemini: A family of highly capable multimodal models. *Technical Report*, Google DeepMind.

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D., Hendricks, L. A., Welbl, J., Clark, A., et al. (2022). Training compute-optimal large language models. In *NeurIPS 2022*.

LeCun, Y. (2022). A path towards autonomous machine intelligence. *OpenReview*, version 0.9.2.

Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. (2023). Emergent world representations: Exploring a sequence model trained on a synthetic task. In *ICLR 2023*.

Merrill, W. and Sabharwal, A. (2023). The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545.

Nakkiran, P., Kaplun, G., Bansal, Y., Yang, T., Barak, B., and Sutskever, I. (2021). Deep double descent: Where bigger models and more data hurt. In *ICLR 2020*.

Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. (2020). Zoom in: An introduction to circuits. *Distill*, <https://distill.pub/2020/circuits/zoom-in>.

OpenAI (2023). technical report. *arXiv preprint arXiv:2303.08774*.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. In *NeurIPS 2022*.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *NeurIPS 2017*.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. (2022). Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *ICLR 2023*.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., and Zhou, D. (2022a). Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS 2022*.

Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., Yogatama, D., Bosma, M., Zhou, D., Miculivicius, D., et al. (2022b). Emergent abilities of large language models. *Transactions on Machine Learning Research*.